

Daily Algebra Drill #1 — Answers & Investigations

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Difference / Sum of Squares & Cubes · Newton's Identity Chain · Disguised Quadratics · Telescoping & Partial Fractions.

Section A Rapid Recognition

(≤15 s each)

Factorise, simplify, or evaluate instantly. No expansion. Pure recognition.

1. Factorise completely: $x^4 - 16$.

Answer: $(x^2 + 4)(x + 2)(x - 2)$

Method: Apply the difference of two squares twice: $(x^2)^2 - 4^2 = (x^2 + 4)(x^2 - 4) = (x^2 + 4)(x + 2)(x - 2)$.

Investigate further: Can you factorise $x^4 + 16$ over the real numbers? Hint: Add and subtract $8x^2$ to complete the square, creating a hidden difference of squares.

2. Simplify: $\frac{a^2 - b^2}{a^2 + 2ab + b^2}$.

Answer: $\frac{a - b}{a + b}$

Method: Factor both numerator and denominator: $\frac{(a - b)(a + b)}{(a + b)^2}$. Cancel out one factor of $(a + b)$.

Investigate further: What if the numerator was $a^3 - b^3$ and the denominator $a^2 + ab + b^2$? Recognise how standard identities pair up to simplify fractions.

3. Without expanding, evaluate: $103^2 - 97^2$.

Answer: 1200

Method: Use difference of two squares: $(103 + 97)(103 - 97) = (200)(6) = 1200$. Always use this for close integers.

Investigate further: Use this concept backward: how can you mentally compute 103×97 ? Treat it as $(100 + 3)(100 - 3) = 10000 - 9$.

4. Factorise: $2x^2 + 7x - 15$.

Answer: $(2x - 3)(x + 5)$

Method: Look for factors of $2 \times -15 = -30$ that sum to 7. These are 10 and -3 . Splitting the middle term and grouping yields the factors.

Investigate further: Use the quadratic formula to find the roots $3/2$ and -5 . Notice how the roots immediately dictate the $(2x - 3)(x + 5)$ structure via the Factor Theorem.

5. Simplify: $\left(x + \frac{1}{x}\right)^2 - \left(x - \frac{1}{x}\right)^2$.

Answer: 4

Method: Use the identity $(A + B)^2 - (A - B)^2 = 4AB$. Here $A = x$ and $B = \frac{1}{x}$, so the result is $4 \cdot x \cdot \frac{1}{x} = 4$.

Investigate further: This algebraic identity has a geometric proof involving the area of rectangles. Try drawing a square of side $(A + B)$ and removing a square of side $(A - B)$.

6. Factorise: $x^3 + 8$.

Answer: $(x + 2)(x^2 - 2x + 4)$

Method: Apply the sum of cubes identity: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ with $a = x$ and $b = 2$.

Investigate further: Why does the quadratic factor $x^2 - 2x + 4$ never have real roots? Check its discriminant and understand how this applies to all sum/difference of cubes.

7. Simplify: $\frac{x^2 - 5x + 6}{x^2 - 4}$.

Answer: $\frac{x - 3}{x + 2}$

Method: Factor the numerator as $(x - 2)(x - 3)$ and the denominator as $(x - 2)(x + 2)$. Cancel the common $(x - 2)$ term.

Investigate further: Sketch the graph of this function. The cancelled $(x - 2)$ factor doesn't just disappear—it creates a "hole" (removable discontinuity) at $x = 2$. What is the y -coordinate of this hole?

8. If $a + b = 5$ and $ab = 3$, find $a^2 + b^2$.

Answer: 19

Method: Use the symmetric polynomial identity: $a^2 + b^2 = (a + b)^2 - 2ab = 25 - 6 = 19$. Never compute a and b explicitly.

Investigate further: Now find $a^3 + b^3$ using the formula $(a + b)^3 - 3ab(a + b)$. You should be able to do this purely mentally.

9. Factorise: $4x^2 - 12x + 9$.

Answer: $(2x - 3)^2$

Method: Recognise the perfect square trinomial structure: $(ax - b)^2 = a^2x^2 - 2abx + b^2$. Here $a = 2$ and $b = 3$.

Investigate further: A quadratic is a perfect square if and only if its discriminant equals zero. Verify that $b^2 - 4ac = 0$ for this expression.

10. Without a calculator, simplify: $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots + \frac{1}{9 \cdot 10}$.

Answer: $\frac{9}{10}$

Method: Use partial fractions: $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. The series telescopes to $(1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) \cdots = 1 - \frac{1}{10} = \frac{9}{10}$.

Investigate further: Generalise this to $\sum_{k=1}^n \frac{1}{k(k+2)}$. The partial fraction split is $\frac{1}{2}(\frac{1}{k} - \frac{1}{k+2})$, which creates a delayed telescoping effect.

Section B Manipulation Drills

(≤50 s each)

Algebraic fractions, rearranging, hidden structure. Resist the urge to expand.

1. Simplify: $\frac{x^3 - 1}{x^2 - 1}$.

Answer: $\frac{x^2 + x + 1}{x + 1}$

Method: Factor the numerator as a difference of cubes: $(x - 1)(x^2 + x + 1)$. Factor the denominator as $(x - 1)(x + 1)$. Cancel $(x - 1)$.

Investigate further: This is $\frac{x^3-1}{x-1} \div \frac{x^2-1}{x-1}$. Recognise the geometric series polynomials: you are effectively dividing $(x^2 + x + 1)$ by $(x + 1)$.

2. Simplify: $\frac{1}{x-1} - \frac{2}{x^2-1}$.

Answer: $\frac{1}{x+1}$

Method: The common denominator is $(x-1)(x+1)$. Multiply the first fraction by $\frac{x+1}{x+1}$. The numerator becomes $(x+1) - 2 = x-1$, which perfectly cancels with the denominator.

Investigate further: Try simplifying $\frac{1}{x-1} + \frac{1}{x+1} - \frac{2}{x^2-1}$. Grouping the terms creatively often reveals the cancellation faster.

3. Make x the subject: $\frac{x+a}{x-a} = k \quad (k \neq 1)$.

Answer: $x = a \frac{k+1}{k-1}$

Method: Cross-multiply: $x+a = k(x-a)$. Expand: $x+a = kx-ka$. Collect x terms: $x-kx = -ka-a$. Factor: $x(1-k) = -a(k+1) \implies x = a \frac{k+1}{k-1}$.

Investigate further: This is a Möbius transformation. Notice that if $f(x) = \frac{x+a}{x-a}$, then the inverse function $f^{-1}(k)$ essentially has the exact same structural form!

4. Simplify: $\frac{a}{b-c} + \frac{b}{c-a} + \frac{c}{a-b}$, given that $a+b+c=0$.

Answer: 0

Method: Since $a+b+c=0$, we have $a = -(b+c)$. Substitute this into the first numerator: $\frac{-(b+c)}{b-c}$.

Do the same for the others. Putting them over a common denominator $(a-b)(b-c)(c-a)$ causes all terms to sum exactly to 0.

Investigate further: The condition $a + b + c = 0$ is famously linked to $a^3 + b^3 + c^3 = 3abc$. Try proving that identity by expanding $(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$.

5. Show that $\frac{x^2 + 3x + 2}{x^2 - x - 2}$ simplifies, and state all values of x for which the original expression is undefined.

Answer: $\frac{x+1}{x-2}$; undefined at $x = 2$ and $x = -1$.

Method: Factor the numerator as $(x+1)(x+2)$ and denominator as $(x-2)(x+1)$. Cancel $(x+1)$. The original fraction is undefined wherever the original denominator is zero.

Investigate further: What is the horizontal asymptote of this rational function? As $x \rightarrow \infty$, the ratio of the leading coefficients dictates the limit.

6. Simplify: $\left(\frac{x^2}{y} - \frac{y^2}{x}\right) \div \left(\frac{x}{y} + 1 + \frac{y}{x}\right)$.

Answer: $x - y$

Method: Write both brackets with common denominator xy .

First bracket: $\frac{x^3 - y^3}{xy}$. Second bracket: $\frac{x^2 + xy + y^2}{xy}$.

The xy denominators cancel, leaving $\frac{x^3 - y^3}{x^2 + xy + y^2}$, which is precisely the formula for $(x - y)$.

Investigate further: What changes if the first bracket is $\frac{x^2}{y} + \frac{y^2}{x}$ instead? How does the second bracket need to adjust to result in $x + y$?

7. Factorise: $a^2(b-c) + b^2(c-a) + c^2(a-b)$.

Answer: $-(a-b)(b-c)(c-a)$ or $(a-b)(c-b)(c-a)$

Method: By the Factor Theorem, if setting $a = b$ makes the expression zero, then $(a - b)$ is a factor. By cyclic symmetry, $(b - c)$ and $(c - a)$ are also factors. To check the sign, expand a test term: the given expression has $+a^2b$, but $(a - b)(b - c)(c - a)$ expands to $-a^2b$, hence the negative sign is required.

Investigate further: This cyclic alternating polynomial is the determinant of a 3×3 Vandermonde matrix. Investigate what a Vandermonde matrix is and why its determinant factors so beautifully.

8. Solve for x : $\frac{3}{x+1} + \frac{4}{x+2} = \frac{11}{(x+1)(x+2)}$.

Answer: $x = 1$

Method: Multiply the entire equation by the common denominator $(x+1)(x+2)$. This leaves a simple linear equation: $3(x+2) + 4(x+1) = 11 \implies 7x + 10 = 11 \implies 7x = 1$.

Investigate further: If the numerators were x instead of constants, you would end up with a quadratic equation. Always check if your final roots cause a division by zero in the original fractions!

9. Simplify: $\frac{2^{n+2} + 2^n}{2^{n+1} + 2^{n-1}}$.

Answer: 2

Method: Factor out the smallest power from both top and bottom.

Top: $2^n(2^2 + 1) = 2^n(5)$.

Bottom: $2^{n-1}(2^2 + 1) = 2^{n-1}(5)$.

The 5 cancels, leaving $2^n/2^{n-1} = 2^1 = 2$.

Investigate further: Can you see that the numerator is exactly 2 times the denominator just by looking at the indices? $(n+2)$ is one higher than $(n+1)$, and n is one higher than $(n-1)$.

10. Given that $p + q = 7$ and $p^3 + q^3 = 133$, find pq .

Answer: $pq = 10$

Method: Use the identity $p^3 + q^3 = (p + q)[(p + q)^2 - 3pq]$.

Substitute the knowns: $133 = 7[49 - 3pq]$. Divide by 7: $19 = 49 - 3pq$.

Rearrange: $3pq = 30 \implies pq = 10$.

Investigate further: Now that you have $p + q = 7$ and $pq = 10$, you can construct the quadratic $x^2 - 7x + 10 = 0$. Its roots are 2 and 5, revealing the underlying values of p and q effortlessly.

Section C Substitution & Structure

(≤90 s each)

Look for symmetry, disguised quadratics, and useful substitutions before computing anything.

1. Solve: $x^4 - 5x^2 + 4 = 0$.

Answer: $x = \pm 1, \pm 2$

Method: This is a disguised quadratic. Let $u = x^2$. The equation becomes $u^2 - 5u + 4 = 0$, which factorises to $(u - 1)(u - 4) = 0$. Hence $x^2 = 1 \implies x = \pm 1$ and $x^2 = 4 \implies x = \pm 2$.

Investigate further: What condition must k satisfy for $x^4 - kx^2 + 4 = 0$ to have exactly four distinct real roots? (Answer: $k > 4$). Analyse the discriminant and root positivity.

2. If $x + \frac{1}{x} = 3$, find $x^3 + \frac{1}{x^3}$.

Answer: 18

Method: Square the given equation: $x^2 + 2 + \frac{1}{x^2} = 9 \implies x^2 + \frac{1}{x^2} = 7$.

Now multiply $(x + \frac{1}{x})(x^2 + \frac{1}{x^2}) = 3 \times 7 = 21$.

Expanding the left gives $x^3 + \frac{1}{x} + x + \frac{1}{x^3} = 21$. Substitute $x + \frac{1}{x} = 3$ to get 18.

Investigate further: Alternatively, memorise the direct identity $t^3 - 3t$ where $t = x + \frac{1}{x}$. Use this to quickly evaluate $x^5 + \frac{1}{x^5}$.

3. Find all real solutions to: $(x^2 - 3x)^2 - 2(x^2 - 3x) - 8 = 0$.

Answer: $x = -1, 1, 2, 4$

Method: Substitute $u = x^2 - 3x$. The equation becomes $u^2 - 2u - 8 = 0$, giving $(u - 4)(u + 2) = 0$.

Case 1: $x^2 - 3x = 4 \implies x^2 - 3x - 4 = 0 \implies (x - 4)(x + 1) = 0$.

Case 2: $x^2 - 3x = -2 \implies x^2 - 3x + 2 = 0 \implies (x - 2)(x - 1) = 0$.

Investigate further: This highlights the power of chunking. Try solving $(x^2 - 3x + 1)^2 - 4(x^2 - 3x + 1) + 3 = 0$ using the exact same strategy.

4. Simplify: $\frac{x^2 + y^2}{xy} - \frac{(x - y)^2}{xy}$.

Answer: 2

Method: Since they share a denominator, combine them: $\frac{x^2 + y^2 - (x^2 - 2xy + y^2)}{xy}$. The squared terms perfectly cancel, leaving $\frac{2xy}{xy} = 2$.

Investigate further: This implies $\frac{x^2 + y^2}{xy} \geq 2$ because $(x - y)^2 \geq 0$ for real numbers. You have just proved a form of the AM-GM inequality!

5. The expression $x^4 + 4y^4$ can be factorised over the integers. Do so.

Answer: $(x^2 + 2y^2 + 2xy)(x^2 + 2y^2 - 2xy)$

Method: Add and subtract $4x^2y^2$ to complete a square:

$$x^4 + 4x^2y^2 + 4y^4 - 4x^2y^2 = (x^2 + 2y^2)^2 - (2xy)^2.$$

Now apply difference of two squares to get the final answer.

Investigate further: This is the famous Sophie Germain Identity. Use it to prove that $n^4 + 4$ is never a prime number for any integer $n > 1$.

6. Solve: $\sqrt{2x + 1} + \sqrt{2x - 1} = t$ for x in terms of t ($t > 0$).

Answer: $x = \frac{t^4 + 4}{8t^2}$

Method: Multiply the given equation by its conjugate: $(\sqrt{2x + 1} + \sqrt{2x - 1})(\sqrt{2x + 1} - \sqrt{2x - 1}) = (2x + 1) - (2x - 1) = 2$.

Thus, the conjugate $\sqrt{2x + 1} - \sqrt{2x - 1} = 2/t$.

Add the original equation and the conjugate: $2\sqrt{2x + 1} = t + 2/t = \frac{t^2 + 2}{t}$.

Square both sides: $4(2x + 1) = \frac{t^4 + 4t^2 + 4}{t^2} \implies 8x + 4 = t^2 + 4 + \frac{4}{t^2} \implies 8x = \frac{t^4 + 4}{t^2}$.

Investigate further: The conjugate trick is extremely powerful for radical equations. Why is squaring the original equation directly $\dots = t^2$ so much messier here?

7. Given that $a + b + c = 0$, evaluate $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$.

Answer: 3

Method: Combine the fractions over the common denominator abc . The numerator becomes $a^3 + b^3 + c^3$. We know that when $a + b + c = 0$, $a^3 + b^3 + c^3 = 3abc$. So the expression is $\frac{3abc}{abc} = 3$.

Investigate further: Investigate how to compute $a^4 + b^4 + c^4$ when $a + b + c = 0$. (Hint: square $a^2 + b^2 + c^2$ and use symmetric polynomials).

8. Solve the system: $x + y = 5$, $x^2 + y^2 = 17$.

Answer: $(x, y) = (1, 4)$ or $(4, 1)$

Method: Use $(x+y)^2 = x^2 + y^2 + 2xy \implies 25 = 17 + 2xy \implies xy = 4$.

Since $x+y = 5$ and $xy = 4$, x and y are the roots of the quadratic $t^2 - 5t + 4 = 0$, which factors as $(t-1)(t-4) = 0$.

Investigate further: Geometrically, you just found the intersection points of the line $y = -x + 5$ and the circle $x^2 + y^2 = 17$. Sketch this to visually confirm the symmetric coordinates.

Section D Challenge

(TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Find the value of

$$\frac{(2025)^3 - (2024)^3 - (2023)^3 + (2022)^3}{(2025)^2 - (2022)^2}.$$

Answer: 2

Method: Let $x = 2022$. The expression is $\frac{(x+3)^3 - (x+2)^3 - (x+1)^3 + x^3}{(x+3)^2 - x^2}$.

Numerator pairing: $[(x+3)^3 - x^3] - [(x+2)^3 + (x+1)^3]$.

Expand safely: $(x^3 + 9x^2 + 27x + 27 - x^3) - (x^3 + 6x^2 + 12x + 8 + x^3 + 3x^2 + 3x + 1)$.

$= (9x^2 + 27x + 27) - (2x^3 + 9x^2 + 15x + 9)$. Wait, the sign on x^3 was positive! Let's pair differently:

$$(x+3)^3 - (x+2)^3 = 3x^2 + 15x + 19.$$

$$x^3 - (x+1)^3 = -3x^2 - 3x - 1.$$

Add them together: $12x + 18$.

Denominator: $(x+3)^2 - x^2 = 6x + 9$. The ratio is exactly $\frac{2(6x+9)}{6x+9} = 2$.

Investigate further: Notice how the x^3 and x^2 terms completely vanish. This means for ANY sequence of four consecutive integers a, b, c, d , the ratio $\frac{d^3 - c^3 - b^3 + a^3}{d^2 - a^2}$ is a constant. Prove it.

2. Let $f(x) = \frac{x}{x+1}$. Define f^n as f composed with itself n times. Find a closed form for $f^n(x)$ and hence compute $f^{2025}(1)$.

Answer: $f^n(x) = \frac{x}{nx+1}$; $f^{2025}(1) = \frac{1}{2026}$

Method: Compute the first few iterates to spot the pattern:

$$f^2(x) = f\left(\frac{x}{x+1}\right) = \frac{\frac{x}{x+1}}{\frac{x}{x+1}+1} = \frac{x}{x+(x+1)} = \frac{x}{2x+1}.$$

$$f^3(x) = \frac{\frac{x}{2x+1}}{\frac{x}{2x+1}+1} = \frac{x}{3x+1}.$$

By induction, $f^n(x) = \frac{x}{nx+1}$. Therefore $f^{2025}(1) = \frac{1}{2025(1)+1} = \frac{1}{2026}$.

Investigate further: Rational functions of the form $\frac{ax+b}{cx+d}$ are Möbius transformations. They can be composed by multiplying 2×2 matrices! Try representing $f(x)$ as a matrix $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ and finding its n -th power.

3. Positive reals a, b, c satisfy $abc = 1$. Simplify:

$$\frac{1}{1+a+ab} + \frac{1}{1+b+bc} + \frac{1}{1+c+ca}.$$

Answer: 1

Method: Multiply the numerator and denominator of the second fraction by a : $\frac{a}{a+ab+abc} = \frac{a}{a+ab+1}$.
Multiply the numerator and denominator of the third fraction by ab : $\frac{ab}{ab+abc+a^2bc} = \frac{ab}{ab+1+a}$.

All three fractions now share the exact same denominator $1 + a + ab$.

The numerators sum to $1 + a + ab$, making the entire expression equal to 1.

Investigate further: Try generalising this trick for 4 variables a, b, c, d where $abcd = 1$. What do the denominators look like?

4. Without solving for x , find the value of $x^2 + \frac{1}{x^2}$ if $x - \frac{1}{x} = \sqrt{5}$. Hence find the exact value of $x^4 + \frac{1}{x^4}$.

Answer: $x^2 + \frac{1}{x^2} = 7$; $x^4 + \frac{1}{x^4} = 47$

Method: Square both sides: $(x - \frac{1}{x})^2 = x^2 - 2 + \frac{1}{x^2} = (\sqrt{5})^2 = 5$.

Add 2: $x^2 + \frac{1}{x^2} = 7$.

Square this new relation: $(x^2 + \frac{1}{x^2})^2 = x^4 + 2 + \frac{1}{x^4} = 7^2 = 49$.

Subtract 2: $x^4 + \frac{1}{x^4} = 47$.

Investigate further: Use this upward-building technique to evaluate $x^3 - \frac{1}{x^3}$ starting from the original given equation. Remember the minus sign!

5. Find all integers n such that $n^2 + 4n + 3$ is a perfect square.

Answer: $n = -1, -3$

Method: Complete the square: $n^2 + 4n + 3 = (n + 2)^2 - 1$.

We want this to equal some perfect integer square m^2 .

$(n + 2)^2 - m^2 = 1$. This is a difference of two squares: $(n + 2 - m)(n + 2 + m) = 1$.

Since we are dealing with integers, the only integer factors of 1 are $(1)(1)$ and $(-1)(-1)$.

This forces $m = 0$, meaning $(n + 2)^2 = 1 \implies n + 2 = \pm 1$.

So $n = -1$ or $n = -3$. (Both yield 0, which is 0^2).

Investigate further: Equations of the form $x^2 - dy^2 = 1$ are known as Pell's Equations. Here $d = 1$ makes it trivial, but what if you needed $n^2 + 4n + 2 = m^2$? Investigating this opens up a beautiful branch of Number Theory.

Top 5 Patterns Today

- Difference / Sum of Squares & Cubes.** $(a^2 - b^2)$, $(a^3 \pm b^3)$ — always factor these before executing any other algebraic operations. It reveals the structure immediately.
- Newton's Identity Chain.** If you know $p + q$ and pq , you can find $p^n + q^n$ by building upward iteratively. You never need to find the explicit (often messy radical) values of p or q .
- Disguised Quadratics.** When an expression has the form $[f(x)]^2 + b \cdot f(x) + c = 0$, substitute $u = f(x)$ immediately. Do not expand it into a higher-degree polynomial.
- Telescoping & Partial Fractions.** Products or sums of consecutive integers almost always telescope. Look for splits like $\frac{1}{n} - \frac{1}{n+1}$ to collapse the middle terms into oblivion.
- Constraint Exploitation.** When a constraint is given (e.g., $a + b + c = 0$ or $abc = 1$), it is *always* there to be used. Use it to substitute and collapse fractions before attempting brute-force common denominators.

Common Traps to Avoid

- 1. Expanding when you should factor.** Expanding $x^4 - 16$ or multiplying out $(x + 1)(x + 3)$ before recognising a structure wastes time and completely obscures the elegant path to the answer.
- 2. Solving for individual variables.** In systems like $p + q = 7, p^3 + q^3 = 133$, trying to solve for p explicitly via substitution is a trap. Always use symmetric polynomial properties to manipulate the block forms.
- 3. Ignoring domain restrictions after cancellation.** In B5, the cancelled factor $(x + 1)$ creates a hole at $x = -1$. Even though it vanishes from the algebra, it remains a restriction on the domain of the original expression.
- 4. Assuming $a^3 + b^3 + c^3 = (a + b + c)^3$ when $a + b + c = 0$.** The correct identity is $a^3 + b^3 + c^3 = 3abc$ when the sum is zero. The cube of the sum is always zero, which is absolutely not the same as the sum of the cubes!
- 5. Missing the elegant substitution in D-level questions.** Before writing any working down, ask: *What is the structure? Can I substitute? Is there a constraint I haven't used?* Every D-level question is designed to have a route under 4 lines.

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

other prep to look at today:
Daily Integral for Calculus and Logic